

Advanced Predictive Analysis

REG NO-23MID0319

NAME-A.dinesh

ABSTRACT:

House price prediction is an important application of machine learning that helps estimate property values accurately based on various housing features. In this project, the California Housing dataset is used to predict house prices using factors such as median income, house age, number of rooms, population, and proximity to the ocean. Before building the models, data preprocessing and exploratory data analysis (EDA) were performed to clean the dataset and understand the relationships between the features. Five regression algorithms—Simple Linear Regression, Multiple Linear Regression, Polynomial Regression, Ridge Regression, and Lasso Regression—were implemented and compared. The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. The model with the best performance was identified based on these evaluation metrics, demonstrating the effectiveness of machine learning techniques for accurate house price prediction.

1. INTRODUCTION

Machine learning is widely used to solve prediction problems in various fields, including real estate. Predicting house prices accurately helps buyers, sellers, and investors make informed decisions and understand the market value of properties.

House prices are influenced by several factors such as median income, house age, number of rooms, population, and location. By analyzing

historical housing data, machine learning models can identify patterns and predict house prices more accurately than manual estimation.

In this project, different regression algorithms are implemented to build a house price prediction model using the California Housing dataset. The performance of each model is evaluated and compared using standard metrics to identify the most accurate prediction model.

2. PROBLEM STATEMENT

House prices are affected by multiple factors, making accurate price estimation a challenging task. Traditional methods of predicting house prices are often time-consuming and may not provide consistent results. Therefore, there is a need for a machine learning-based prediction system that can analyze housing data, identify patterns, and estimate house prices accurately and efficiently

3. OBJECTIVES

The main objectives of this project are:

- To load and analyze the California Housing dataset.
- To preprocess and clean the dataset for model training.
- To perform Exploratory Data Analysis (EDA) to understand the data and identify patterns.
- To develop house price prediction models using different regression algorithms.
- To evaluate and compare the performance of the regression models using standard evaluation metrics.
- To identify the most accurate and efficient model for predicting house prices

4. DATASET DESCRIPTION

The California Housing dataset contains housing and demographic information collected from different districts in California. Each record represents a housing block and includes several features that influence house prices. The dataset is commonly used for developing and evaluating machine learning models for house price prediction.

Input Features

- Longitude
- Latitude
- Housing Median Age
- Total Rooms
- Total Bedrooms
- Population
- Households
- Median Income
- Ocean Proximity

Target Variable

- **Median House Value (median_house_value)**

5. Methodology

The project was carried out through the following steps:

Step 1: Data Collection

The California Housing dataset was imported from a CSV file and loaded into the Python environment for analysis.

Step 2: Data Preprocessing

The dataset was cleaned by checking for missing values, duplicate records, and incorrect data types. Numerical features were standardized, and the categorical feature (**ocean_proximity**) was converted into numerical form using One-Hot Encoding.

Step 3: Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset and identify relationships between variables. The following analyses and visualizations were used:

- Dataset summary and statistical information
- Missing value analysis
- Correlation heatmap
- Distribution plots
- Scatter plots

Step 4: Data Splitting

The dataset was divided into training and testing sets, with **80%** of the data used for training the models and **20%** used for testing their performance.

Step 5: Model Development

Five regression algorithms were implemented to predict house prices:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Ridge Regression
- Lasso Regression

6. EVALUATION METRICS

Model Evaluation

The performance of the regression models was evaluated using the following metrics:

Mean Absolute Error (MAE):

Measures the average absolute difference between the actual and predicted house prices. A lower MAE indicates better prediction accuracy.

Root Mean Squared Error (RMSE):

Measures the average magnitude of prediction errors by giving greater

importance to larger errors. A lower RMSE indicates a more accurate model.

R² Score (Coefficient of Determination):

Indicates how well the regression model explains the variation in house prices. An R² value closer to 1 represents better model performance.

7. RESULTS AND DISCUSSION

All five regression models were trained and evaluated using the California Housing dataset. The performance of each model was compared using MAE, RMSE, and R² Score. Multiple Linear Regression provided better prediction accuracy by utilizing all relevant input features, while Polynomial Regression effectively captured non-linear relationships within the data. Ridge Regression and Lasso Regression improved model generalization by reducing overfitting through regularization. Based on the evaluation results, the models were compared to identify the most accurate and reliable algorithm for house price prediction.

8. CONCLUSION

This project successfully developed a machine learning-based house price prediction model using the California Housing dataset. The dataset was preprocessed, analyzed, and used to train five different regression algorithms. The models were evaluated and compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score to determine their prediction performance. The results showed that comparing multiple regression techniques helps identify the most accurate model for house price prediction. Overall, this project demonstrates the effectiveness of machine learning in

estimating house prices and highlights its potential to support better decision-making in the real estate industry.

9. FUTURE ENHANCEMENTS

The project can be further enhanced by incorporating larger and more up-to-date housing datasets to improve prediction accuracy. Advanced machine learning algorithms such as Random Forest, XGBoost, and Gradient Boosting can be implemented to achieve better performance. The model can also be deployed as a web application using Flask or Streamlit, allowing users to predict house prices through a simple interface. In addition, integrating real-time housing market data and enabling automatic model retraining with newly available datasets can help maintain the accuracy and reliability of the prediction system over time.

10. References

1. Géron, A. *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. 'Reilly Media.
2. *Scikit-Learn Official Documentation*. Available at
3. *Python Official Documentation*.
4. *California Housing Dataset Documentation*. Available through Scikit-Learn Datasets.
5. Research articles and journals on House Price Prediction using Machine Learning.

Code:

```
#23mid0319
```

```
import numpy as np
```

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.compose import ColumnTransformer

from sklearn.pipeline import Pipeline

from sklearn.impute import SimpleImputer

from sklearn.preprocessing import StandardScaler, OneHotEncoder,
PolynomialFeatures

from sklearn.linear_model import LinearRegression, Ridge

from sklearn.metrics import mean_absolute_error,
mean_squared_error, r2_score


# Load Dataset

url = "https://raw.githubusercontent.com/ageron/handson-ml2/master/
datasets/housing/housing.csv"

df = pd.read_csv(url)


# Split Features and Target

X = df.drop("median_house_value", axis=1)

y = df["median_house_value"]


# Train-Test Split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

)

Preprocessing

```
num_cols = X_train.select_dtypes(include=np.number).columns
```

```
cat_cols = X_train.select_dtypes(exclude=np.number).columns
```

```
preprocess = ColumnTransformer([
```

```
    ("num", Pipeline([
```

```
        ("imputer", SimpleImputer(strategy="median")),
```

```
        ("scaler", StandardScaler())
```

```
    ]), num_cols),
```

```
    ("cat", Pipeline([
```

```
        ("imputer", SimpleImputer(strategy="most_frequent")),
```

```
        ("onehot", OneHotEncoder(handle_unknown="ignore"))
```

```
    ]), cat_cols)
```

```
])
```

Simple Linear Regression

```
simple_model = LinearRegression()
```

```
simple_model.fit(X_train[["median_income"]], y_train)
```

```
simple_pred = simple_model.predict(X_test[["median_income"]])
```



```
# Multiple Linear Regression
```

```
multiple_model = Pipeline([  
    ("preprocess", preprocess),  
    ("model", LinearRegression())  
])
```

```
multiple_model.fit(X_train, y_train)
```

```
multiple_pred = multiple_model.predict(X_test)
```

```
# Polynomial Regression
```

```
poly_features =  
["median_income", "housing_median_age", "total_rooms"]
```

```
poly_model = Pipeline([  
    ("imputer", SimpleImputer(strategy="median")),  
    ("poly", PolynomialFeatures(degree=2)),  
    ("scaler", StandardScaler()),  
    ("model", LinearRegression())  
])
```

```
poly_model.fit(X_train[poly_features], y_train)
```

```
poly_pred = poly_model.predict(X_test[poly_features])
```

```
# Ridge Regression
```

```
ridge_model = Pipeline([  
    ("preprocess", preprocess),  
    ("model", Ridge(alpha=1.0))  
])
```

```
ridge_model.fit(X_train, y_train)
```

```
ridge_pred = ridge_model.predict(X_test)
```

```
# Store Models
```

```
results = {  
    "Simple Linear Regression": simple_pred,  
    "Multiple Linear Regression": multiple_pred,  
    "Polynomial Regression": poly_pred,  
    "Ridge Regression": ridge_pred  
}
```

```
# Evaluation + Graphs
```

```
for name, pred in results.items():
```

```
    mae = mean_absolute_error(y_test, pred)
```

```
rmse = np.sqrt(mean_squared_error(y_test, pred))
```

```
r2 = r2_score(y_test, pred)
```

```
print("="*60)
```

```
print(name)
```

```
print("="*60)
```

```
print("MAE :", round(mae,2))
```

```
print("RMSE :", round(rmse,2))
```

```
print("R2  :", round(r2,4))
```

```
plt.figure(figsize=(6,5))
```

```
plt.scatter(y_test, pred, alpha=0.5)
```

```
plt.plot(
```

```
    [y_test.min(), y_test.max()],
```

```
    [y_test.min(), y_test.max()],
```

```
    color="red",
```

```
    linewidth=2
```

```
)
```

```
plt.title(name)
```

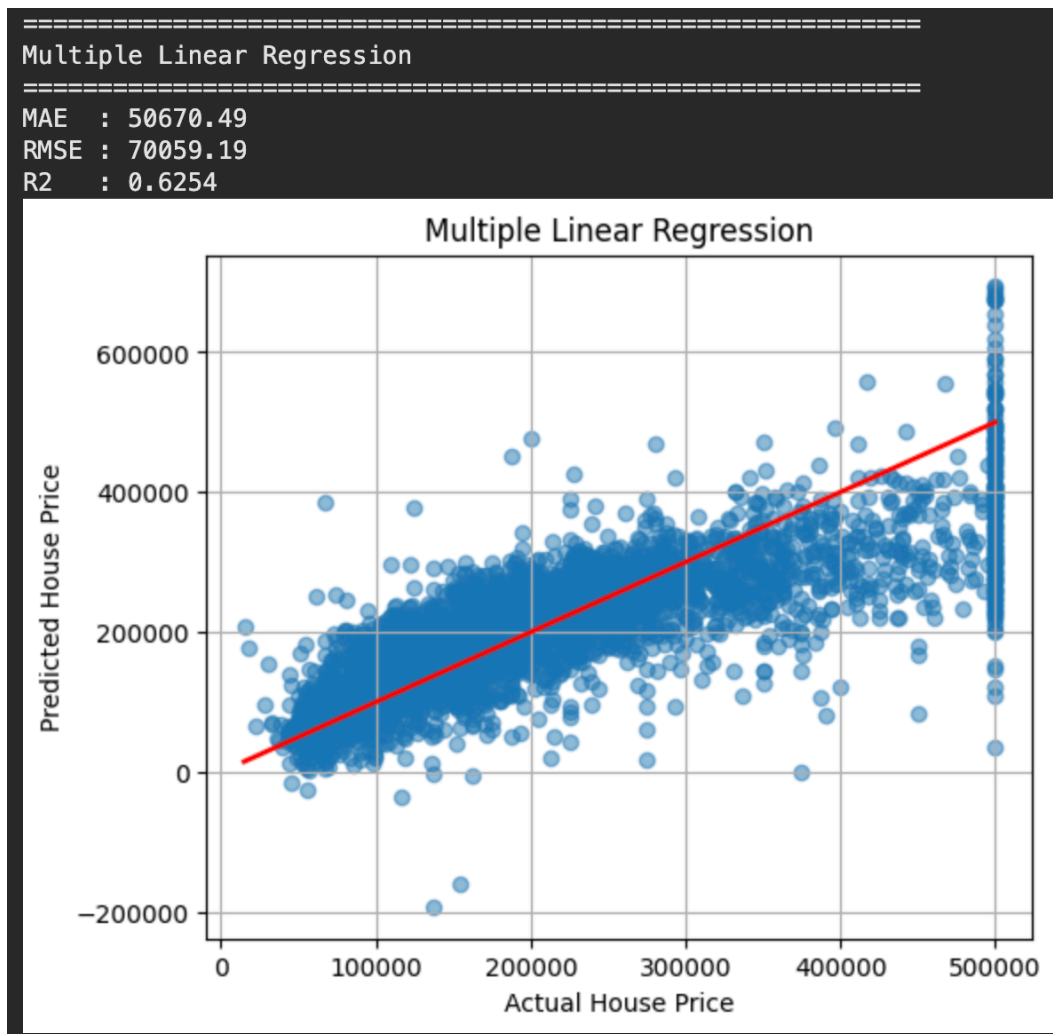
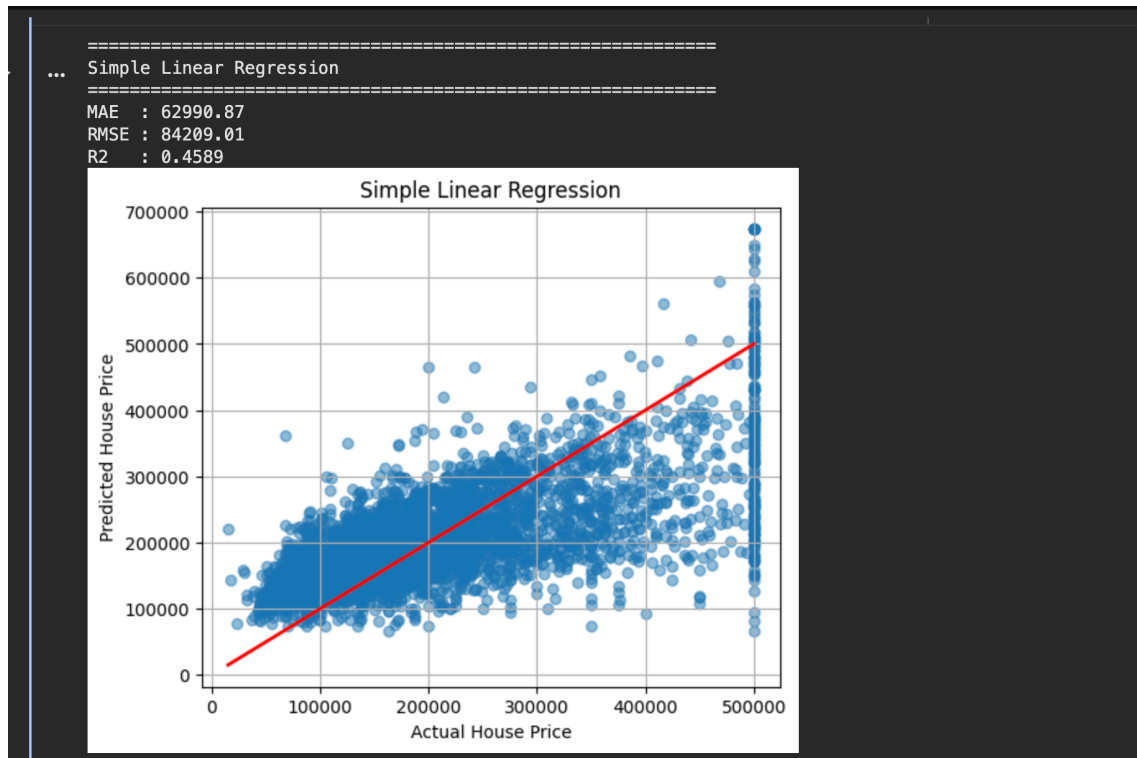
```
plt.xlabel("Actual House Price")
```

```
plt.ylabel("Predicted House Price")
```

```
plt.grid(True)
```

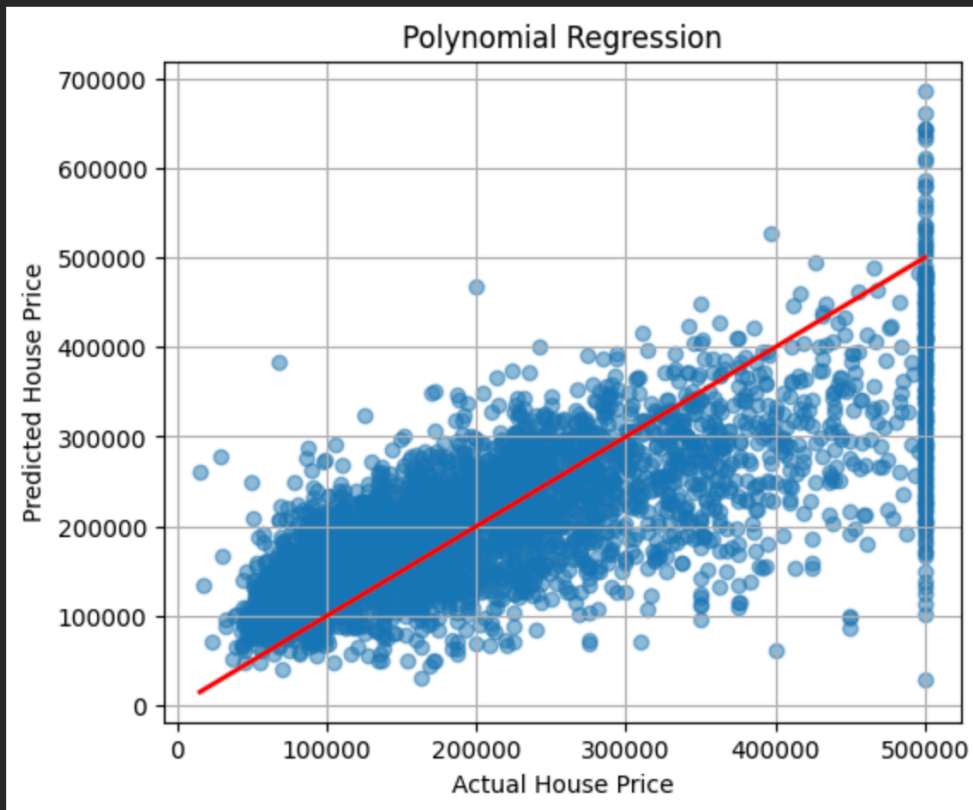
plt.show()

Output:



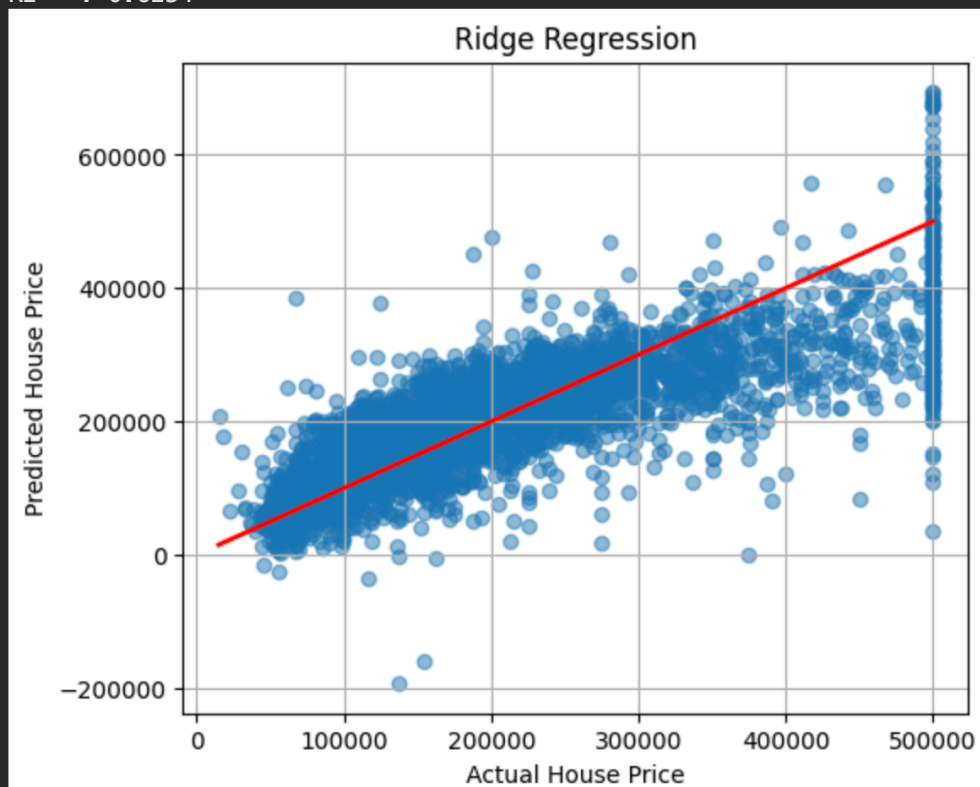
Polynomial Regression

MAE : 58853.91
RMSE : 79152.15
R2 : 0.5219



Ridge Regression

MAE : 50676.92
RMSE : 70066.02
R2 : 0.6254



Github link:

[https://github.com/dineshala/advance-predictive-analysis/
blob/main/data/23MID0319_report%20.pdf](https://github.com/dineshala/advance-predictive-analysis/blob/main/data/23MID0319_report%20.pdf)